Verbal overshadowing in taste recognition ability

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Verbal overshadowing (VO) refers to impaired recognition of a stimulus after it has been verbally described. This effect was first identified in face recognition (Schooler & Engstler-Schooler, 1990) and subsequently extended to a wide range of visual stimuli, including abstract figures and maps. VO has also been observed outside the visual domain, including olfaction (Cao et al., 2024). A common feature of these stimuli is their resistance to precise verbal description. Gustation is even more lexically impoverished than vision in both English (Winter, 2016) and Chinese (Chen et al., 2019), yet VO in taste recognition remains underexplored. Early work by Melcher and Schooler (1996) suggested a VO effect among wine drinkers but not among non-drinkers or wine experts, whereas Hughson and Boakes (2009) reported the opposite pattern, a verbal facilitation effect. These inconsistencies may stem from differing criteria for defining drinking frequency and expertise. To address this gap, the present study incorporated a rigorous pre-test to ensure comparable gustatory functioning across participants while also assessing their verbalisation ability and confidence ratings, thereby controlling for key sources of variability that may have contributed to earlier mixed findings. The present study aimed to advance VO research by shifting from the well-studied visual domain to the far less charted interaction between gustatory memory and language. Across three experiments, participants identified target tastes from sets of four food samples (one target, three distractors; Fig. 1).

Experiment 1 tested 40 native Chinese speakers, with 20 completing a verbal description of the target taste before recognition and 20 completing a filler task (Fig. 2). Recognition accuracy was significantly lower in the Verbal group than in the Control group ($\beta = -1.63$, $SE = 0.48$, $t = -3.41$, $p = .001$), demonstrating a robust VO effect for taste (Fig. 3).

Experiment 2 examined whether explicitly alerting participants to the possibility of VO would mitigate its effect. Twenty participants in the Verbal Two-choice group were informed that their verbal descriptions might differ from their later taste memory (Fig. 2). Nonetheless, recognition accuracy remained significantly lower than in the Control group ($\beta = -1.05$, $SE = 0.48$, $t = -2.16$, $p = .034$) (Fig. 3), indicating that awareness alone does not attenuate VO.

Experiment 3 tested whether both awareness of VO and active suppression of description-based representations could reduce its impact. Twenty participants in the Verbal Ignore group were instructed to disregard any information derived from their descriptions and rely solely on taste memory (Fig. 2). Again, accuracy was significantly lower than in the Control group ($\beta = -1.26$, $SE = 0.53$, $t = -2.39$, $p = .0197$) (Fig. 3), suggesting that even active suppression does not eliminate verbal overshadowing.

Discussion: Together, these findings provide convergent evidence that verbalization reliably impairs gustatory recognition and that neither increased metacognitive awareness nor deliberate suppression moderates this effect. The results are discussed in relation to five candidate mechanisms, including recoding interference, transfer-inappropriate processing shifts, retrieval-based interference, criterion-shift and label-feedback accounts. The results resonate strongly with the focus of this year's HSP conference and can catalyse fresh debates about on how language and thought interact.

Fig. 1. Taste stimuli used across the three experiments.

	Target	Distractors		
Set 1	Apricot	Peach	Apple	Cherry
Set 2	Kiwifruit	Pineapple	Blueberry	Grape

Fig. 2. Flow chart for the four conditions. The red star refers to the instruction of unveiling VOE before the recognition task.

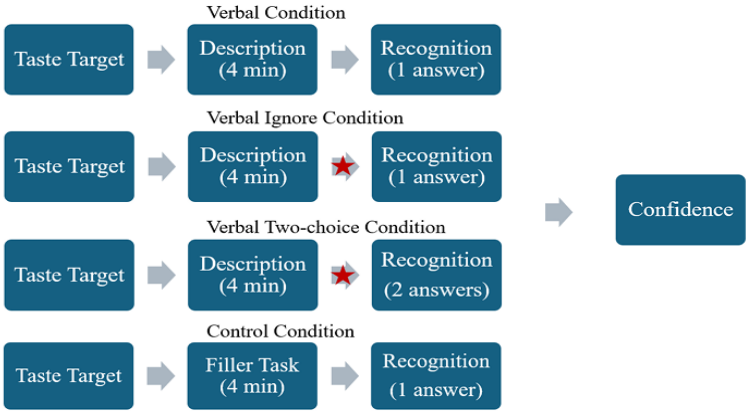
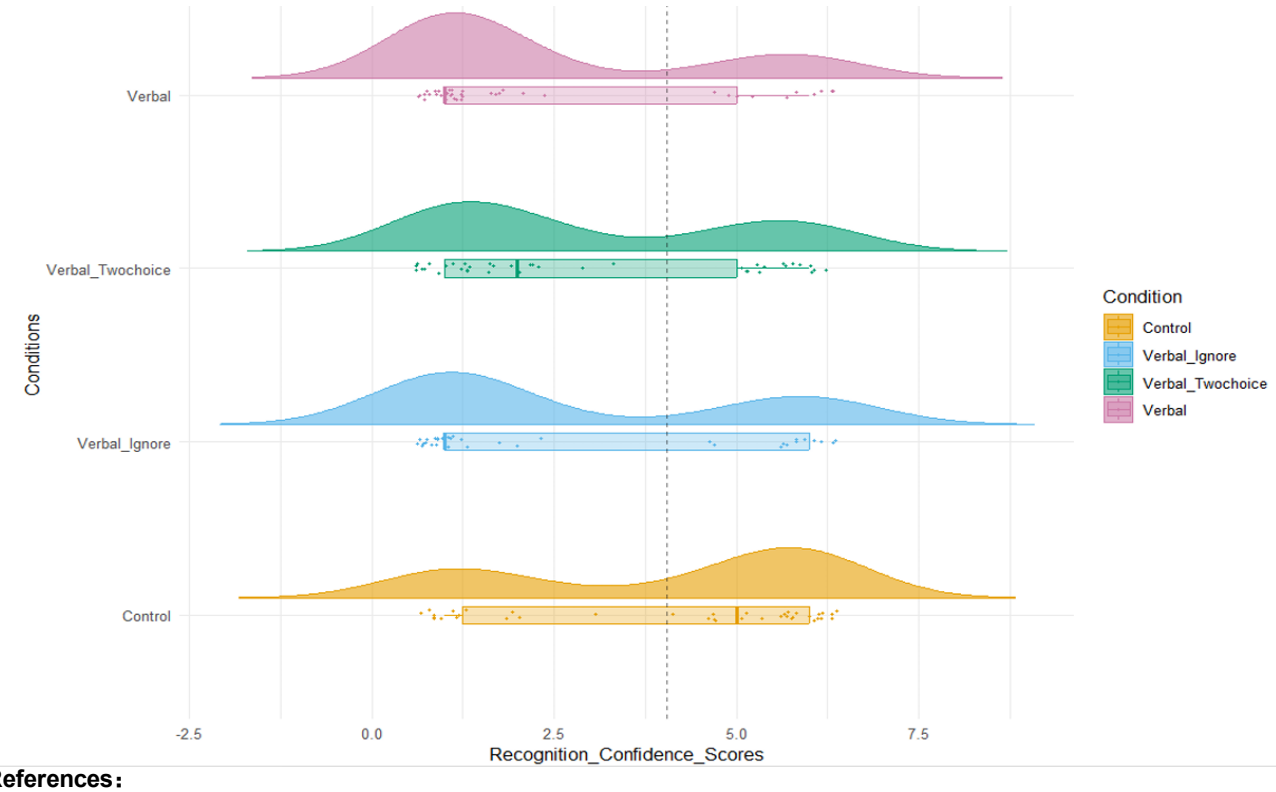


Fig. 3. Distribution of composite scores across conditions, including verbal (Exp. 1), verbal two-choice (Exp. 2), verbal ignore (Exp. 3), and control group. The dashed line marks the mean of the Control group.



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